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EXPERIMENT 13

Enterprise limitations case study

Aim

Analyze where a low code library design may be insufficient for a complex enterprise scenario and propose an appropriate architecture.

Case scenario

A hypothetical university network has 20 branches, one million historical loans, RFID kiosks, intermittent connectivity, central identity, inter-library transfers and peak simultaneous circulation. These figures are scenario assumptions, not measured Creator limits or an actual institution’s data.

Procedure

1. Translate the scenario into explicit acceptance targets: acceptable response time, permitted downtime, data residency, audit retention, concurrent issue safety, offline behavior and recovery time. Obtain stakeholder agreement instead of choosing targets after testing.

2. Map each requirement to a native capability, an extension, or an unresolved constraint. Examine current quotas, API throttling, workflow execution limits, integration availability and licensing in the intended plan.

3. Design proof-of-concept tests for the highest risks: concurrent allocation of one copy, peak report load, delayed synchronization, identity revocation and an interrupted multi-service operation.

4. Compare an all-Creator design, a hybrid design and a custom system using weighted business criteria. Estimate total ownership effort and migration costs with explicit assumptions.

5. Write a recommendation, unresolved risks and exit strategy. Export data with stable IDs, document mappings, and identify workflows that would need to be rewritten if leaving the platform.

Implementation and verification

Risk	Consequence	Mitigation to evaluate
Concurrent allocation	Two borrowers receive one copy.	Authoritative transactional allocation service; no competing direct write path.
Offline kiosks	Conflicting or stale availability.	Online-only final issue or explicit reservation and reconciliation rules.
Large reporting load	Slow operations or quotas exceeded.	Measured queries, pagination, archival and analytical replica if justified.
Cross-service updates	Creator and external state disagree.	Idempotency keys, outbox, retries and reconciliation.
Vendor dependency	Cost or portability constraints.	Contract review, data export drills and documented interfaces.

Case analysis and recommendation

A reasonable hypothesis is to keep catalogue administration and staff dashboards in Creator while using a proven transactional service for high-contention circulation. Validate this choice with measured evidence: added infrastructure and reconciliation costs may outweigh the benefit for a small library. Offline writes cannot be assumed to preserve a global one-copy rule.

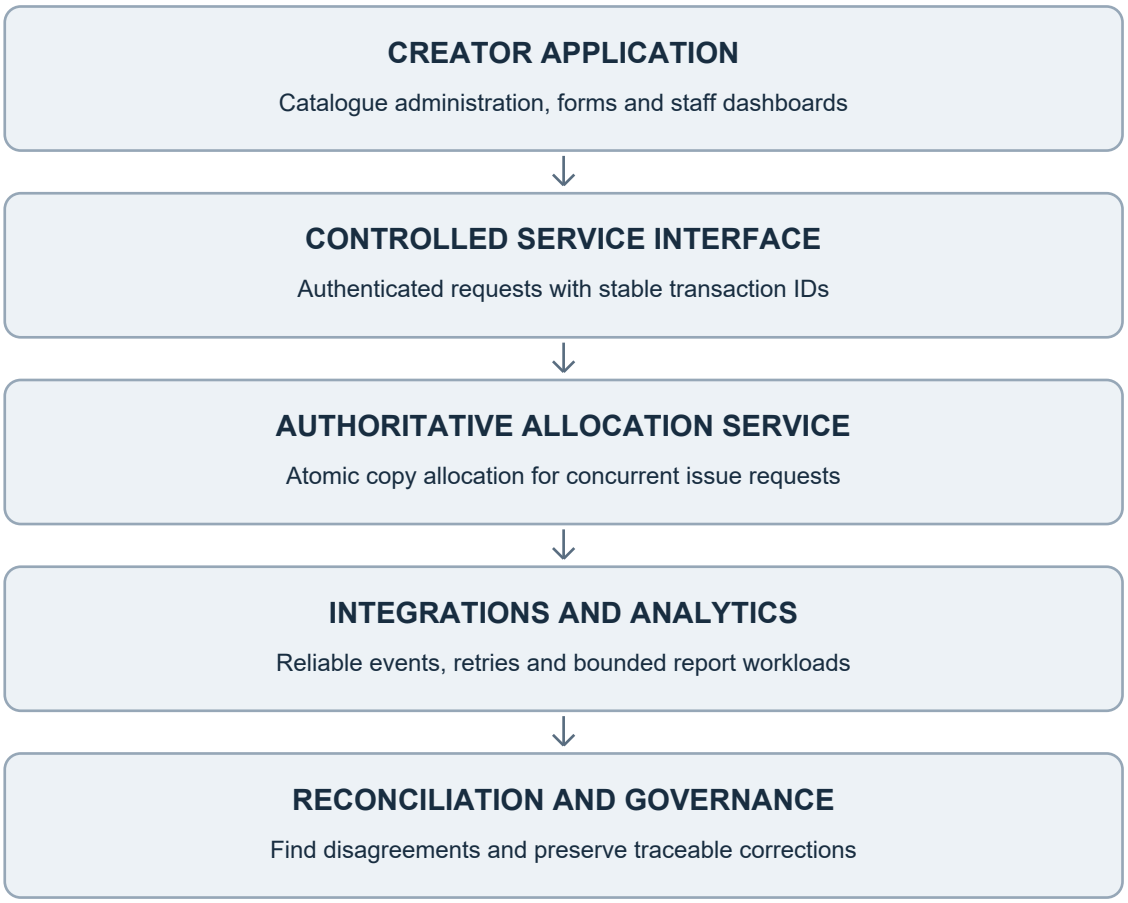
Implementation and interpretation

The hybrid design is a proposal for the hypothetical enterprise case. It must not introduce two independent authorities for book availability. If both Creator and an external service can issue a copy directly, the original race condition can remain unresolved.

EXPERIMENT 13 VISUAL RECORD

Application images and explanation

Illustrative process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

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Result

Expected result and verification criteria. E13 contains a defensible recommendation, explicit assumptions and a testable risk register. It avoids unsupported claims that low code cannot scale or that it automatically meets every enterprise requirement.